INT304: Network Security and Protocols – Lab 1: Introduction to Network Security Concepts

Objectives

- Understand fundamental concepts of network security.
- Identify different types of threats and vulnerabilities in network environments.
- Learn about security protocols and their roles in safeguarding network communications.

Lab Activities

1. **Overview of Network Security**
    - Begin by reading the provided materials on the importance of network security. Focus on why organizations must prioritize securing their networks.
    - Discuss various types of network attacks, including:
        - **Denial of Service (DoS) and Distributed Denial of Service (DDoS)**: Flooding a network to make it unavailable.
        - **Man-in-the-Middle (MitM)**: Intercepting communication between two parties.
        - **Phishing**: Deceiving individuals into providing sensitive information.
2. **Identifying Threats and Vulnerabilities**
    - Conduct a group exercise where each group is given a hypothetical network scenario (e.g., a small business, a university campus).
    - Identify potential threats in the given scenarios, discussing what types of attacks could be launched and the vulnerabilities present in the network architecture.
3. **Understanding Security Protocols**
    - Review key security protocols and their functions:
        - **SSL/TLS (Secure Sockets Layer/Transport Layer Security)**: Protects data transmitted over the internet, ensuring that communication remains private and secure.
        - **IPSec (Internet Protocol Security)**: A suite of protocols designed to secure Internet Protocol (IP) communications through encryption and authentication.

        - **SSH (Secure Shell)**: A protocol for secure remote login and other secure network services.
        - **HTTPS (Hypertext Transfer Protocol Secure)**: An extension of HTTP, using SSL/TLS to secure communications over a computer network.
    - Discuss how each protocol operates and in what situations they are typically used.
4. **Hands-On Activity**
    - Open a web browser and navigate to a secure website (e.g., https://www.example.com).
    - Click on the padlock icon in the address bar to view the SSL/TLS certificate details.
        - Record the certificate issuer, validity period, and encryption strength used for the connection.
    - Discuss findings with your peers about the significance of having secure connections in everyday internet use.
5. **Group Discussion**
    - Facilitate a discussion on the importance of network security in modern businesses.
    - Encourage participants to share insights from the hands-on activity and how analyzing SSL/TLS certificates can impact security practices.

Deliverables

- Compile a brief report (1-2 pages) summarizing your findings from the lab activities. Include:
    - Key threats and vulnerabilities identified during the group exercise.
    - A summary of each discussed security protocol.
    - Insights gained from analyzing the SSL/TLS certificate.

Deadline

- The report should be submitted by the specified due date, which will be communicated in class.

INT304: Network Security and Protocols – Lab 2: Network Security Policies and Risk Management

Objectives

- Understand the importance of network security policies.
- Learn how to conduct a risk assessment.
- Develop skills to create effective security policies for an organization.

Lab Activities

1. **Introduction to Network Security Policies**
    - Start with a reading assignment on network security policies and their significance in safeguarding organizational assets.
    - Discuss the key components of an effective security policy, including:
        - **Purpose and Scope**: What the policy aims to protect.
        - **User Responsibilities**: Expectations and acceptable use of network resources.
        - **Incident Response Plan**: Steps to take when a security breach occurs.
        - **Compliance**: Adherence to relevant laws and regulations.
2. **Conducting a Risk Assessment**
    - Break into small groups and select a hypothetical organization (e.g., a healthcare provider, a financial institution).
    - Each group should identify potential threats and vulnerabilities specific to their chosen organization.
    - Conduct a risk assessment using the following steps:
        - **Asset Identification**: List key assets that require protection (e.g., customer data, intellectual property).
        - **Threat Identification**: Identify potential threats to each asset (e.g., cyberattacks, natural disasters).
        - **Vulnerability Assessment**: Analyze weaknesses that could be exploited by threats.
        - **Impact Analysis**: Evaluate the potential impact of each threat on the organization.
        - **Risk Rating**: Assign a risk rating (high, medium, low) based on likelihood and impact.

3. **Developing Security Policies**
    - Using the findings from the risk assessment, each group will draft a preliminary security policy tailored to their organization. Key elements to include:
        - **Access Control Policies**: Guidelines for user access and authentication.
        - **Data Protection Policies**: Measures to safeguard sensitive information.
        - **Incident Response Procedures**: Steps to be taken in the event of a security incident.
        - **Training and Awareness Programs**: Plans for educating employees about security best practices.
4. **Presentation of Policies**
    - Each group will present their risk assessment and proposed security policy to the class.
    - Encourage peer feedback and discussion on the strengths and weaknesses of each policy.
5. **Reflection and Discussion**
    - Hold a class discussion on the importance of adapting security policies to changing threats and organizational needs.
    - Discuss how organizations can regularly review and update their security policies to remain effective.

Deliverables

- Each group will submit a comprehensive report (3-5 pages) that includes:
    - The completed risk assessment for the chosen organization.
    - The drafted security policy with all key components.
    - A summary of feedback received during the presentation.

Deadline

- The report should be submitted by the specified due date, which will be communicated in class.